Constraints on the solutions to the NBP

Def: Euler's Theorem on Homogeneous Functions

A function $\underline{F(\vec{x})}$, $(F \in \mathbb{R},\ \vec{x} \in \mathbb{R}^n)$ is homogeneous of degree k if

$$F(\alpha \vec{x}) = \alpha^k F(\vec{x}) \text{ for all } \alpha > 0.$$

If a function is homog. of deg. k, then

$$\boxed{\vec{x} \cdot \frac{\partial F}{\partial \vec{x}} = k F(\vec{x})}$$

Apply this to $\mathcal{U} \Rightarrow$ is homog. of degree -1 : $\vec{r} \cdot \frac{\partial \mathcal{U}(\vec{r})}{\partial \vec{r}} = -\mathcal{U}$

$T \Rightarrow$ " " " " 2 : $\vec{v} \cdot \frac{\partial T}{\partial \vec{v}} = 2T$

I_p " " " " 2 : $\vec{r} \cdot \frac{\partial I_p}{\partial \vec{r}} = 2 I_p$

Triangle Inequality : Given $\vec{a}, \vec{b}$, $\underbrace{|\vec{a} \cdot \vec{b}|}_{\vec{a} \cdot \vec{b}} \leq |\vec{a}|\,|\vec{b}|$

$|\vec{a} \times \vec{b}| \leq |\vec{a}| \cdot |\vec{b}|$

Cauchy Inequality:

$$\left(\sum_{k=1}^{n} a_k b_k \right)^2 \leq \left(\sum_{k=1}^{n} a_k^2 \right) \cdot \left(\sum_{k=1}^{n} b_k^2 \right)$$

Using these defs we can find the following results.

Inertial Frame (Assume $\vec{R}_c = 0$)

Lagrange - Jacobi Identity:

$E = T + \mathcal{U}$

Given $I_p = \sum_i m_i r_i^2$, consider

$$\dot{I}_p = 2 \sum_i m_i \vec{r}_i \cdot \vec{v}_i$$

$$\ddot{I}_p = 2 \sum_i m_i \left[\vec{V}_i \cdot \vec{V}_i + r_i \cdot \ddot{\vec{r}}_i \right]$$

$$= 2 \sum_i m_i V_i^2 + 2 \sum \vec{r}_i \cdot \underbrace{(m_i \ddot{\vec{r}}_i)}_{\left(-\frac{\partial \mathcal{U}}{\partial \vec{r}_i}\right)}$$

$$= 4T - 2 \underbrace{\sum_i \vec{r}_i \cdot \frac{\partial \mathcal{U}}{\partial \vec{r}_i}}_{-\mathcal{U}}$$

$$\Rightarrow \boxed{\begin{aligned} \ddot{I}_p &= 4T + 2\mathcal{U} \\ &= 2T + 2E \\ &= 4E - 2\mathcal{U} \end{aligned}}$$

Implications

(I) A system cannot remain stationary (assuming $N \geq 2$, $\mathcal{U} \neq 0$)

If $T = 0$, the system is stationary

Then $\ddot{I}_p = 2\mathcal{U} \neq 0$, meaning that the relative distances (w/in I_p) must change, leading to $T > 0$.

(II) A system with total energy $E > 0$ must have at least one body escape.

$\ddot{I}_p = 2T + 2E \geq 2E > 0$, thus $\boxed{\ddot{I}_p > 0, \text{ also } I_p > 0}$

Thus as $t \to \infty$ $I_p \to \infty$, and at least 1 body is not bounded (different than "escape").

III Virial Thm

If an N-body system is "bound from above & below",

then on average $\ddot{I}_p \to 0$.

$$\ddot{I}_p = 4T + 2\mathcal{U}$$

$$\boxed{\lim_{\tau\to\infty} \frac{1}{\tau}\int_0^\tau \ddot{I}_p \, dt \to 0}$$

$\Rightarrow$

$$\left.\begin{aligned} \overline{T} &= \lim_{\tau\to\infty} \frac{1}{\tau}\int_0^\tau T\,dt \\ \overline{\mathcal{U}} &= \lim_{\tau\to\infty} \frac{1}{\tau}\int_0^\tau \mathcal{U}\,dt \end{aligned}\right\} \text{ are defined}$$

Then $\boxed{\overline{T} = -E, \quad \overline{\mathcal{U}} = 2E, \quad \overline{\mathcal{U}} = -2\overline{T}}$ Note, $E < 0$

Any Behavior of the NBP?

Example: 2 body Problem

$$m_1\vec{r}_1 + m_2\vec{r}_2 = 0 = m_1\vec{v}_1 + m_2\vec{v}_2$$

$$I_p = m_1 r_1^2 + m_2 r_2^2$$

$$m_1\dot{\vec{v}}_1 = -\frac{\partial \mathcal{U}}{\partial \vec{r}_1} \quad \& \quad m_2\dot{\vec{v}}_2 = -\frac{\partial \mathcal{U}}{\partial \vec{r}_2}$$

$$T = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$$

$$\mathcal{U} = \frac{-G m_1 m_2}{|\vec{r}_{12}|}$$

$$\dot{I}_p = 2m_1\vec{r}_1\cdot\vec{v}_1 + 2m_2\vec{r}_2\cdot\vec{v}_2$$

$$\ddot{I}_p = \underbrace{2m_1v_1^2 + 2m_2v_2^2}_{4\cdot T} + 2m_1\vec{r}_1\cdot\dot{\vec{v}}_1 + 2m_2\vec{r}_2\cdot\dot{\vec{v}}_2 = 4T + 2\mathcal{U} \cdots$$

$$- 2\underbrace{\left[\vec{r}_1\cdot\frac{\partial\mathcal{U}}{\partial\vec{r}_1} + \vec{r}_2\cdot\frac{\partial\mathcal{U}}{\partial\vec{r}_2}\right]}_{-\mathcal{U}}$$

$$\boxed{I_p = \frac{m_1m_2}{m_1+m_2}\underbrace{(\vec{r}_1-\vec{r}_2)}_{\vec{r}}\cdot\underbrace{(\vec{r}_1-\vec{r}_2)}_{\vec{r}} = \frac{m_1m_2}{m_1+m_2}r^2}$$

IF $\underline{E<0}$, $H \neq 0$, then $\boxed{\underset{\text{periapsis}}{q} \le r \le \underset{\text{apoapsis}}{Q}}$

$$dM = (1-e\cos\bar{E})\,d\bar{E}$$

$I_p = \frac{m_1 m_2}{m_1+m_2} r^2$ is bnd from above & below.

$\lim_{\tau\to\infty} \frac{1}{\tau}\int \ddot{I}_p\,d\tau \to 0$; $\bar{T}$ & $\bar{\mathcal{U}}$ are defined.....

$$\bar{T} = \frac{1}{2}\,\frac{1}{2\pi}\int_0^{2\pi} \frac{m_1 m_2}{m_1+m_2} v^2\,dM = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}\int_0^{2\pi} \frac{\mu}{a}\,\frac{(1+e\cos\bar{E})}{(1-e\cos\bar{E})}\,dM$$

$$= \frac{1}{2}\,\frac{m_1 m_2}{m_1+m_2}\,\frac{\mu}{a}\,\underbrace{\frac{1}{2\pi}\int_0^{2\pi}(1+e\cos\bar{E})\,d\bar{E}}_{1} =$$

$$\boxed{\bar{T} = \frac{1}{2}\,\frac{m_1 m_2}{\cancel{m_1+m_2}}\,\frac{G\cancel{(m_1+m_2)}}{a}}$$

$r = a(1-e\cos\bar{E})$

$$\bar{\mathcal{U}} = -G m_1 m_2 \frac{1}{2\pi}\int_0^{2\pi} \frac{dM}{r} = -G m_1 m_2 \frac{1}{2\pi}\int_0^{2\pi} \frac{d\bar{E}}{a} = \frac{-G m_1 m_2}{a}; \quad \bar{\mathcal{U}} = -2\bar{T} \checkmark$$

$$\boxed{\bar{E} = \bar{T} + \bar{\mathcal{U}} = -\frac{G m_1 m_2}{2a}}$$

Apply Cauchy Inequality to the Angular Momentum

$$\vec{H} = \sum_i m_i \vec{r}_i \times \vec{v}_i$$

$$H^2 = \left(\sum_i m_i \vec{r}_i \times \vec{v}_i\right) \cdot \left(\sum_i m_i \vec{r}_i \times \vec{v}_i\right) \le \underbrace{\left[\sum_i m_i r_i v_i\right]^2}_{\text{Triangle Ineq}} \le \underbrace{\overset{I_p}{\left(\sum_i m_i r_i^2\right)} \overset{2T}{\left(\sum_i m_i v_i^2\right)}}_{\text{Cauchy}}$$

$$\boxed{H^2 \le I_p 2T}$$

$$T = E - \mathcal{U}$$

$$\boxed{H^2 \le 2 I_p (E - \mathcal{U})}$$

$$\Downarrow$$

$$\boxed{\mathcal{E}_p = \frac{H^2}{2I_p} + \mathcal{U} \le E}$$

$$\boxed{\mathcal{E}_p = \frac{H^2}{2I_p} + \mathcal{U} \quad \text{the amended potential}}$$

Is only a function of relative positions.

Stronger version of the Amended potential.

$$\vec{H} = \sum_i m_i \vec{r}_i \times \vec{v}_i \,, \quad \hat{H} = \frac{\vec{H}}{H}$$

$$\hat{H} \cdot \Big(\overbrace{- \sum_i m_i \tilde{r}_i \cdot \tilde{r}_i}^{\bar{\bar{I}}} \Big) \cdot \hat{H}$$

$2T$

$$H = \hat{H} \cdot \vec{H} = \Big(\sum_i m_i \vec{r}_i \times \vec{v}_i \Big) \cdot \hat{H} = \sum_i m_i \, \hat{H} \cdot \big(\underbrace{\vec{r}_i \times \vec{v}_i}_{\tilde{r}_i \cdot \vec{v}_i} \big)$$

$$H^2 = \Big(\quad \Big)\Big(\quad \Big) \le \Big[\sum_i m_i \, |\hat{H} \cdot \tilde{r}_i| \, |\vec{v}_i| \Big]^2 \le \Big(\sum_i m_i (\hat{H} \cdot \tilde{r}_i)^2 \Big) \Big(\sum_i m_i v_i^2 \Big)$$

$$H^2 \le 2 \underbrace{(\hat{H} \cdot \bar{\bar{I}} \cdot \hat{H})}_{I_H} T = 2 I_H T$$

$T = E - \mathcal{U}$

Can show that $\boxed{I_H \le I_p}$

So $\quad H^2 \le 2 I_H T \le 2 I_p T \Rightarrow$

$$\underbrace{\mathcal{E}_p = \frac{H^2}{2I_p} + U}_{} \le \underbrace{\mathcal{E}_H = \frac{H^2}{2I_H} + U}_{\text{Sharper: Will play an important role later}} \le E$$

Sundman's Inequality

$$\ddot{I}_p = 4E - 2U = 2T + 2E \ge 2T + \frac{\cancel{2}H^2}{\cancel{2}I_p} + 2U = 2E + \frac{H^2}{I_p}$$

$$\boxed{\ddot{I}_p \ge 2E + \frac{H^2}{I_p}}$$

Theory of total collapse

Total Collapse of NBP occurs if **all** $r_{ij} \to 0$ simultaneously ($I_p \to 0$)

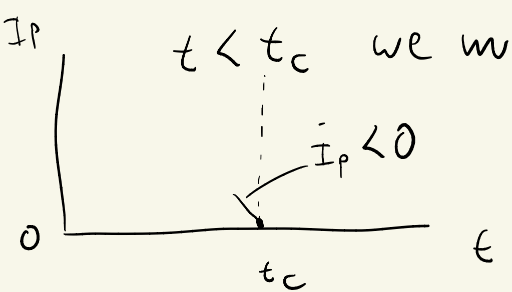
① If Tot. Coll. occurs, it happens at a finite time.

Note $\ddot{I}_p = 4E - 2U$, if T.C. occurs $E = \text{const}$

$\boxed{\text{but } -2U \to \infty}$ & $\ddot{I}_p \to \infty$, Can show this cannot happen as $t \to \pm\infty$.

② If T.C. occurs, $H = 0$.

If a T.C. occurs at time t_c, then at some time $t < t_c$ we must have $\dot{I}_p < 0$.



$$\left(\ddot{I}_p \geq 2E + \frac{H^2}{I_p} \right) \quad \text{if} \quad \dot{I}_p < 0 \quad \text{befn} \quad t = t_c$$

$$\boxed{-\dot{I}_p \ddot{I}_p \geq -\dot{I}_p \left[2E + \frac{H^2}{I_p} \right] = -2E\dot{I}_p - H^2 \frac{\dot{I}_p}{I_p}}$$

$$\int \left(-\frac{d}{dt}\left(\frac{1}{2}\dot{I}_p^2\right) \geq -2E\frac{dI_p}{dt} - H^2 \frac{\frac{dI_p}{dt}}{I_p} \right) dt$$

$$\int_t^{t_c} -d\left(\frac{1}{2}\dot{I}_p^2\right) \geq \int_t^{t_c} -2E\, dI_p - \int_t^{t_c} H^2 \frac{dI_p}{I_p}$$

$$\boxed{-\frac{1}{2}\dot{I}_p^2(t) + K \geq -2E I_p(t) + H^2 \ln\left(\frac{1}{I_p(t)}\right)} \Rightarrow \boxed{H^2 \leq \frac{2EI_p(t) + K}{\ln\left(1/I_p(t)\right)}}$$

$$t \longrightarrow t_c$$

$$K \geq$$

$$H^2 \le \frac{2E\,I_p(t) + K}{\ln\left(1/I_p(t)\right)}$$

; If $I_p(t) \to 0$

$H^2 \to 0$

but H^2 is a constant and if $H^2 \neq 0 \Rightarrow$ get a bound on how small $I_p(t)$ can be.

and then we can't have $I_p \to 0$.